Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 25: Approved Solutions

Counterexamples

Covers Chartrand & Zhang, Section 3.5

Status of this set

Ungraded practice. Solutions released with it. Attempt everything first.

Part I: How to Practice This Material

Check the hypothesis first

A counterexample to $P(x) \Rightarrow Q(x)$ needs $P(a)$ **true** and $Q(a)$ **false**. Verify the hypothesis *before* the conclusion — an element failing the hypothesis proves nothing.

Where counterexamples hide

Zero. Negative numbers. Fractions, when the domain is $\mathbb{R}$. The empty set. Equal elements, when a claim assumes distinct ones. Try these before anything else.

Three mistakes to expect

1. **Offering an element that fails the hypothesis.**
2. **Describing a counterexample instead of naming one.** "Some negative number" is not an element.
3. **Disproving a biconditional without saying which direction failed.**

Part II: Practice Problems

Section A — Disprove

1. Disprove each by giving a counterexample, and verify both halves.

(a) Let $n \in \mathbb{Z}$. If n^2 is divisible by 4, then n is divisible by 4.

(b) Let $x \in \mathbb{R}$. If $x^2 = 9$, then $x = 3$.

(c) Let $n \in \mathbb{Z}$. If n is prime, then $n + 2$ is prime.

Approved solution

(a) $n = 2$. Hypothesis: $n^2 = 4$, divisible by 4 — true. Conclusion: 2 is not divisible by 4 — false. ✓
(b) $x = -3$. Hypothesis: $(-3)^2 = 9$ — true. Conclusion: $-3 \neq 3$ — false. ✓
(c) $n = 7$. Hypothesis: 7 is prime — true. Conclusion: $9 = 3 \cdot 3$ is not prime — false. ✓(Also $n = 2$, since 4 is not prime.)

Part (b) is the domain trap. Over the positive reals the claim holds; over $\mathbb{R}$ it fails on a negative. Same lesson as the midterm's Problem 7.

2. For each attempted disproof, decide whether it is valid. If not, say why and supply a valid counterexample.

(a) Claim: *if n is odd, then n^2 is odd.* Offered: $n = 2$, since 4 is even.

(b) Claim: *if $n > 0$, then $n^2 > n$.* Offered: $n = 1$.

Approved solution

(a) Invalid. $n = 2$ is even, so the hypothesis fails and the implication is true in that row. In fact the claim is *true* — there is no counterexample to supply. That is Theorem 3.25.
(b) Valid. Hypothesis: $1 > 0$ — true. Conclusion: $1^2 > 1$ is $1 > 1$ — false. So $n = 1$ is a genuine counterexample.

(a) is the lesson. An attempted disproof of a true claim will always fail the hypothesis somewhere — and noticing that is how you discover the claim was true all along.

Section B — Biconditionals and sets

3. Disprove: let $m, n \in \mathbb{Z}$; then $m + n$ is odd if and only if m and n are both odd. Say which direction fails.

Approved solution

Both directions fail, but one counterexample per direction is clearest.
Reverse direction (if both odd, then $m+n$ odd): $m = n = 1$ gives $m+n = 2$, even. So the reverse implication fails.
Forward direction (if $m+n$ odd, then both odd): $m = 1$, $n = 2$ gives $m+n = 3$, odd, but n is not odd. So this fails too.
Either counterexample alone disproves the biconditional.

The true statement is that $m + n$ is odd if and only if m and n have opposite parity — Practice Set 24, Problem 3. A disproof is most useful when it points to the correct version.

4. Disprove: for all sets A, B, C, $\quad A \cup (B \cap C) = (A \cup B) \cap C$.

Approved solution

Take $A = \{1\}$, $B = C = \emptyset$. Then $B \cap C = \emptyset$, so the left side is $A = \{1\}$. But $A \cup B = \{1\}$ and $(A \cup B) \cap C = \{1\} \cap \emptyset = \emptyset$.
The element 1 is in the left side and not the right, so the sets differ.

The correct identity is $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ — the distributive law. The false version drops the A from the second factor. Choosing C empty exposed that immediately.

Section C — Prove or disprove

5. Decide whether each is true or false. Prove it or give a counterexample.

(a) Let $n \in \mathbb{Z}$. If n^2 is even, then n is even.

(b) Let $x \in \mathbb{R}$. If $x > 1$, then $x^2 > x$.

(c) Let $n \in \mathbb{Z}$. If n is even, then n^2 is divisible by 8.

Approved solution

(a) True. This is Theorem 3.24; the contrapositive (if n is odd then n^2 is odd) was proved in Lecture 23.
(b) True. Let $x > 1$. Multiplying both sides of $x > 1$ by the positive number x gives $x^2 > x$. ■
(c) False. $n = 2$: even, but $n^2 = 4$ is not divisible by 8.
(It *is* true that n^2 is divisible by 4 when n is even, since $n = 2k$ gives $n^2 = 4k^2$.)

Test before committing. Trying $n = 2, 4, 6$ in (c) gives $4, 16, 36$, and 4 fails at once. Trying $x = 2, 3$ in (b) gives no failure, which suggests looking for a proof.

Section D — Writing

6. In five to eight complete sentences, explain why an element that fails the hypothesis of an implication cannot serve as a counterexample. Use the truth table for $\Rightarrow$ and one specific example.

Approved solution

Model answer. An implication $P \Rightarrow Q$ is false in exactly one row of its truth table: the row where P is true and Q is false. A counterexample to the universal statement $\forall x,\ P(x) \Rightarrow Q(x)$ must therefore be an element that lands in that row, since only there is the implication actually false. An element for which $P(x)$ is false lands in one of the rows where the implication is true regardless of Q, so it confirms the statement at that element rather than refuting it. For example, to disprove "if n is divisible by 4, then n is divisible by 8," the value $n = 3$ is useless: it is not divisible by 4, so the implication is vacuously true there. The value $n = 4$ works, because it satisfies the hypothesis and fails the conclusion. Checking the hypothesis first is the fastest way to avoid offering an element that proves nothing.

Full credit requires the single false row and a concrete example of an element that fails the hypothesis. An answer that only states the rule earns about half.

Section E — Challenge (optional)

7. Disprove: for every positive integer n, the number $n^2 + n + 41$ is prime.

Approved solution

Take $n = 40$. Then $n^2 + n + 41 = 1600 + 40 + 41 = 1681 = 41^2$, which is not prime.
(Also $n = 41$: $41^2 + 41 + 41 = 41 \cdot 43$.)

This is the Week 1 polynomial in disguise. Homework 1 used $n^2 - n + 41$, prime for $n = 0, \ldots, 40$

and failing at 41. This version is the same polynomial shifted by one, so it fails one step earlier. Forty consecutive confirming cases, then a failure. It was the strongest argument in Week 1 for why checking cases is not proof, and it is still the best counterexample in the course.

Instructor notes

- **Problem 2(a) is the key item.** An attempted disproof of a *true* claim always fails the hypothesis. Noticing that is how students learn the claim was true.
- **Problem 3** rewards pointing to the correct statement. A disproof that names the true version is worth more than one that does not.
- **Problem 4** uses the empty set to break a set identity instantly. Edge cases first.
- **Problem 7** closes a loop from Homework 1. Worth mentioning even to students who skip it.
- Reserved for Homework 10: Section 3.5 Exercises 10, 12, 14, 16 (even). Odd ones have printed answers and are self-check only.